1. Determine whether two given strings are anagrams of each other.

2. Convert a non-negative decimal integer into its binary string representation.

3. Determine whether a given year satisfies the leap year rules.

4. Find the intersection of two sorted arrays and return the common elements.

5. Remove punctuation from a sentence.

6. Find the length of a given string.

7. Split a sentence into individual words and store in a list.

8. Sort a list of words in alphabetical order.

9. Given an unsorted list of integers and a target value T, find the exact position of T in the list, or report that it doesn't exist.

10. Scan a paragraph of text and count how many times each individual vowel (a, e, i, o, u) appears.

11. Given a list of IP addresses that accessed a server today, remove all duplicates and return only the unique IP addresses.

12. Given a square grid of numbers ($N \times N$), calculate the sum of the numbers that lie on the main diagonal from top-left to bottom-right.

13. Reverse a string without using built-in reverse functions.

14. Find the sum of all even numbers in a list.

15. Find the second largest in an array.

16. Find GCD of two numbers using Euclidean algorithm.

17. Compute the sum and average of a list of numbers.

18. Count the frequency of each character in a string.

19. Swap two variables without using a temporary third variable.

20. Determine all divisors of a given positive integer.

21. Create a process that determines whether a number is even or odd.

22. Design a procedure that computes the factorial of a positive integer.

23. Write a process that converts numerical scores into grade categories such as Pass, Merit, or Distinction.

24. Create a process that checks whether a password contains at least 8 characters, one number, and one uppercase letter.

25. Find the length of the longest substring without repeating characters in a given string.

26. Perform level-order traversal on a binary tree and return node values grouped by depth.

27. Rotate a square 2D matrix 90 degrees clockwise in-place.

28. For a fixed window size sliding across an array, return the maximum value within each window position.

29. Build a text classification model to categorize text into predefined categories.

30. Create a topic modelling system to extract underlying topics from a large corpus.

31. Develop a co-reference resolution system to identify relationships between pronouns and nouns.

32. Build a question answering system to answer user queries based on a given text.

33. Implement a method to find the exact square root of a non-negative integer using binary search, rounded down to the nearest integer, without using built-in math functions.

34. Find all elements that appear in two separate unsorted lists, ensuring the result contains no duplicates.

35. Design a memory buffer that holds a fixed number of items. When it runs out of space, it must automatically discard the item that hasn't been accessed for the longest time.

36. An ordered list of numbers was rotated at an unknown pivot point (e.g., [4,5,6,1,2,3]). Find the index of a target element in this list in logarithmic time.

37. Implement a stack using an array (push/pop/peek).

38. Find the first non-repeating character in a string.

39. Detect cycle in a singly linked list (Floyd’s cycle detection).

40. Implement a queue using two stacks.

41. Solve the N Queens Problem for N arbitrary board size N.

42. Find all pairs in an array that sum to a given target value.

43. Solve the Tower of Hanoi problem for N discs.

44. Find the contiguous subarray with the maximum sum (Kadane's algorithm).

45. Create instructions for adding two matrices of equal size.

46. Develop a process for multiplying two matrices.

47. Create an algorithm that updates inventory quantities after purchases and deliveries.

48. Develop a recursive process that searches through nested folders to locate a specific file type.

49. Create a pathfinding solution that guides a user through a maze using graph traversal.

50. Develop algorithms enabling autonomous drones to coordinate movement while avoiding collisions.